

Computer Architecture

Assignment 4

Due: Next section. (one day before if you submit by email)

Notes: This assignment is individual assignment, every student should complete by himself.

1. (5 points) True or false

- a. In Load/store architecture, operate instructions operate only on registers. ✓
- b. RISC ISA usually implement many address modes. +
- c. x86 architecture store each instruction in 32 bits. ✗
- d. 0-address machine usually called queue machine. ✗
- e. 0-address machine is better than 2-address machine. ✗
- f. All ISAs provide double linked list as a primitive data type ✗
- g. ISAs with fewer number of data types are easier for programmers ✗
- h. Registers are a set of small storage places within the processor. ✓
- i. Most ISAs provide registers with a size range between 1 MB and 5 MB. ✗
- j. More registers can decrease processor performance and by preventing data locality. ✓
- k. Instructions can be divided into different classes include operate instruction, data movement instructions, control flow instructions. ✓

2. (5 points) The following assembly programs implement Fibonacci sequence in x86 and ARM:

a. x86	b. ARM
<pre> .data n byte 5 .code main PROC mov eax, 1 movzx ecx, n startLoop: mul ecx loop startLoop main ENDP END main </pre>	<pre> .global main .text main: MOV R2, #5 ;Init the character MOV R1, #1 loop: MOV R3,R1 MUL R1,R3,R2 SUB R2, R2, #1 ;Increment it CMP R2, #1 ;Check the limit BGE loop </pre>

- a. Refer to this [x86 timing manual](#) and calculate number of cycles for program “a” if compiled with Intel Core 2 -L32 and AMD-k10
- b. Refer to this [ARM timing manual](#) and calculate number of cycles for program “b” if compiled with Cortex-M3.
- c. Conclude the difference between the results of question 4.a and 4.b.